

Remote-Controlled R/F System

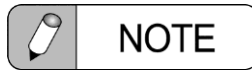
FLEXAVISION F4

Light Viewer

(Version 1.0.0.06)

Instruction Manual

Read the instruction manual thoroughly before you use the product. Keep this instruction manual for future reference.



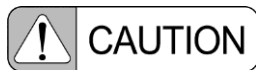
Precautions and prohibitions seen through this manual are classified as follows:



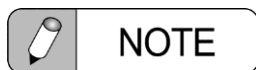
States a direct danger that may cause death or serious injury if it is not avoided.



States an indirect or potential danger that may cause death or serious injury if it is not avoided.



States a danger that may cause slight or medium injury or may cause damage in equipment or fire if it is not avoided.



States information that helps to use the system correctly.

DISCLAIMER

- This software is not intended to be used for diagnostic purposes.
- Before using the software, you have responsibility not to violate any legal or regulatory provisions in your country or region in all perspective including the healthcare and imports/exports regulations. Shimadzu, Canon and the dealers do not owe any liability concerning the software and its use.

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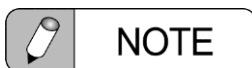
Revision History

<i>Document Version</i>	<i>Date of Issue</i>	<i>Description</i>
<i>First</i>	<i>Apr. 10, 2021</i>	<i>First issue</i>

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This manual explains the operations for CXDI Light Viewer. The Light Viewer is the software to view the recorded images in the disk. It is not necessary to install the Light Viewer into the computer to be used.



- In this manual, measurement results and notes to be placed on the image are called "object"
- In this manual, screens for Windows 7 are used as an example.

Precautions

- Do not use the images stored in the disk for any purpose other than viewing them with the Light Viewer.
- The Light Viewer may not work properly when viewing images captured with other equipment.
- This software is not intended to be used for diagnostic purposes.

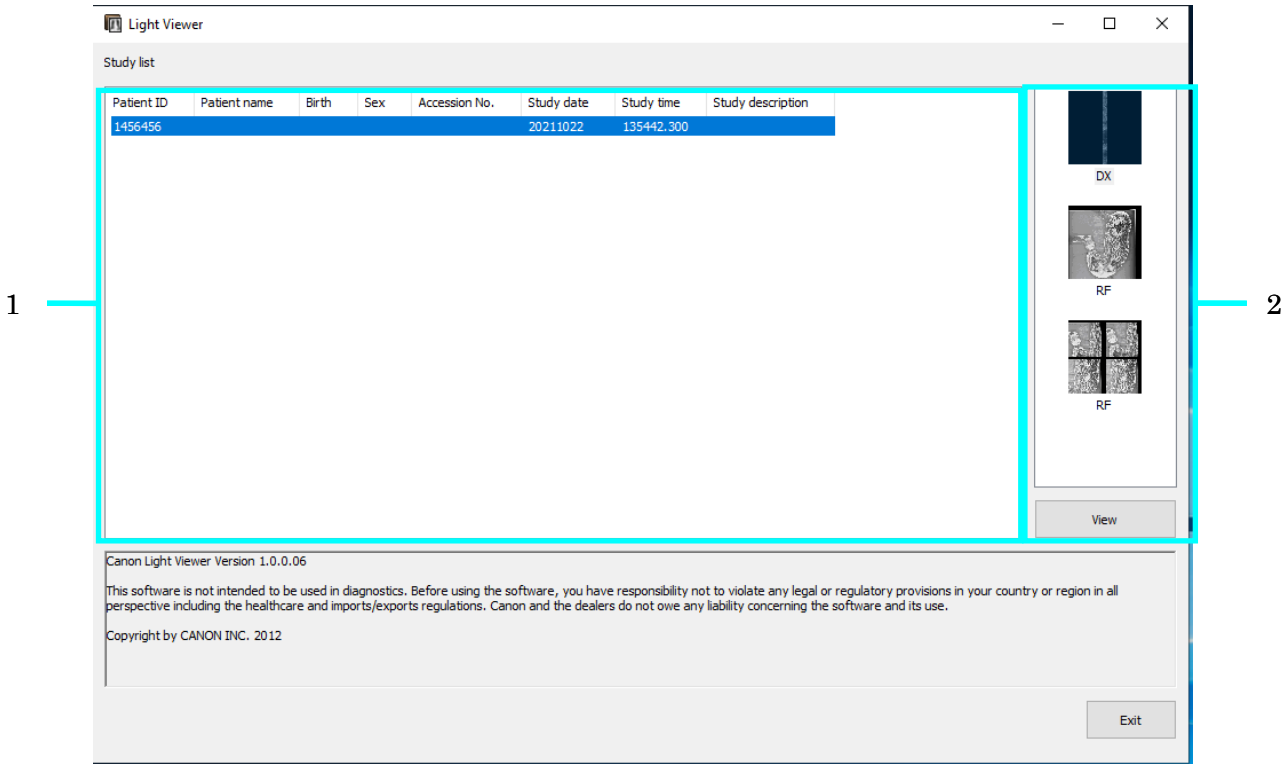
Operating Environment

Hardware or software	Requirements
OS	Microsoft Windows 10 / 7 / Vista / XP SP3 32-bit and 64-bit versions are available.
Display	Screen resolution: SVGA (800×600) or more

- For further requirements (CPU, RAM or HDD capacity etc.), follow the specified system requirements by the OS. For details, refer to the operation manual of your computer.
- The following settings and other installed software may interfere with proper operation of the Light Viewer.
 - Your computer does not meet the recommended specifications.
 - Anti-virus software
 - Other high-memory requiring software
- Configuring the environment as described above does not guarantee proper operation.

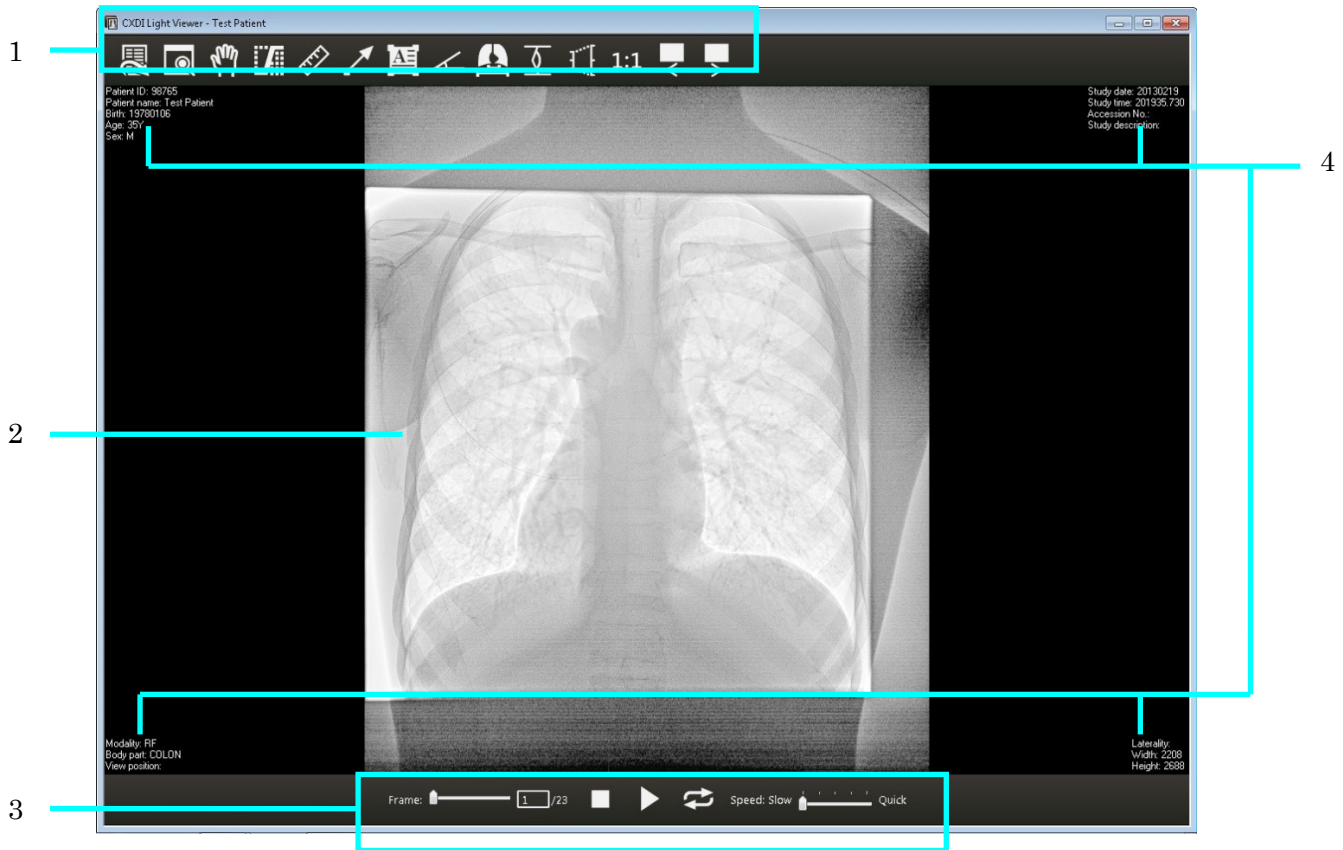
Part names of the screens

Study list screen



- 1 Study list
- 2 Image list of the selected study

Viewer screen



- 1 Tool buttons
- 2 Image view pane
- 3 Dynamic image playback tool
- 4 DICOM tag value

Upper right: Study date, Study time, Accession NO., Study description

Bottom right: Laterality, Width, Height

Upper left: Patient ID, Patient name, Birth, Age, Sex

Bottom left: Modality, Body part, View position

Starting and shutting down the Light Viewer

- 1** Double click [LV.EXE] in the disk.
The Light Viewer starts and study list screen is displayed.
- 2** Select the study.
The image list is displayed.
- 3** Double click an image.
Or click an image and click [View].
The viewer screen is displayed.
- 4** Click .
- 5** Click [Exit].
The Light Viewer shuts down.



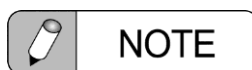
NOTE

- Returning the screen from viewer to the study list screen resets the change of the image display, arrangement of objects, and image processing results.
- The result of the gamma adjustment is saved even if the Light Viewer is shut down.

Changing the display of the image

Zooming

- 1 Click .
- 2 Point the cursor to an image and drag it.
To enlarge the image, drag it to up.
To reduce the image, drag it to down.

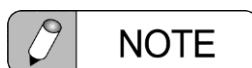


- The magnification can be selected from [Zoom] in the menu displayed by right-clicking.
- To return the magnification to the initial settings, select [Reset zoom] from [Reset] in the menu displayed by right-clicking.

Panning Images

Moving an image allows the users to view target parts of the image at the certain position.

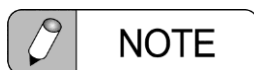
- 1 Click .
- 2 Point the cursor to an image and drag it.



- To pan the image, right-click and drag it.
- To return the image position to the initial settings, select [Reset panning] from [Reset] in the menu displayed by right-clicking.

Switching to pixel-to-pixel display

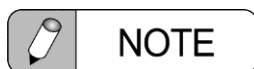
Click . Clicking the button toggles the image between overview and pixel-to-pixel display.



Right-clicking menu also enables you to toggle the display. Select [Pixel to Pixel].

Toggling images

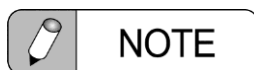
Click  or . The previous or next image in the image list is displayed.



Toggling the image resets the change of the image display, arrangement of objects, and image processing results.

Placing measurement results or notes on images

Measurement results and notes can be placed on images.



- Up to 16 objects can be arranged.
- Objects can be moved by dragging.
- To modify the object, drag an end point of it.
- To select two or more objects, drag the area that included the target objects.
- Objects are deleted by returning to the study list screen or displaying another image. To save the objects, print the image (see P. 14).
- Objects can be also arranged by selecting a menu displayed by right-clicking. Select an item from [Draw object].

Measuring the distance between two specific points.

The corrected value in X-Ray magnification is displayed by measuring the distance between two specific points.

1 Click 

2 Click two specified points.

A line object and the measure distance between the points appear. Measured value, X-Ray magnification and corrected value is displayed.

Placing the arrow objects

1 Click .

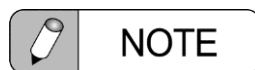
2 Click two specified points.

An arrow object appears.

Placing notes

Notes can be created and placed on an image.

- 1 Click .
- 2 Click two diagonally opposite points.
Text region is displayed.
- 3 Enter notes and then click outside the note region.



Up to 512 characters can be entered. To modify the note, double-click the note region.

Measuring the angle of a specific part

Measures the angle of a specific part and displays the measured value on the image.

- 1 Click .
- 2 Click three specified points.
An angle object and measured value are displayed.

Measuring the CTR

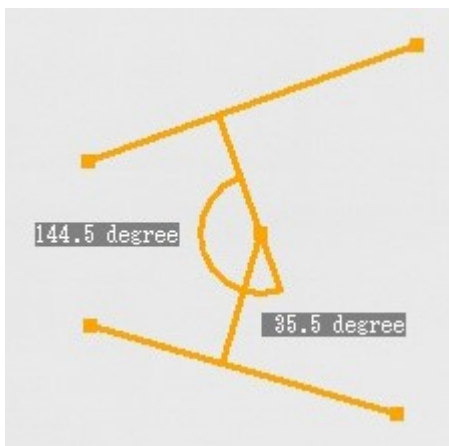
Measures the CTR (cardio thoracic ratio) and displays the measured value on the image.

- 1 Click .
- 2 Click two specified points along the transverse diameter of the thorax.
A CTR object and measured value are displayed.
- 3 Drag two inside points along the transverse diameter of the heart.
The measured value is modified.

Measuring the Cobb angle

Measures the Cobb angle and displays the measured value on the image.

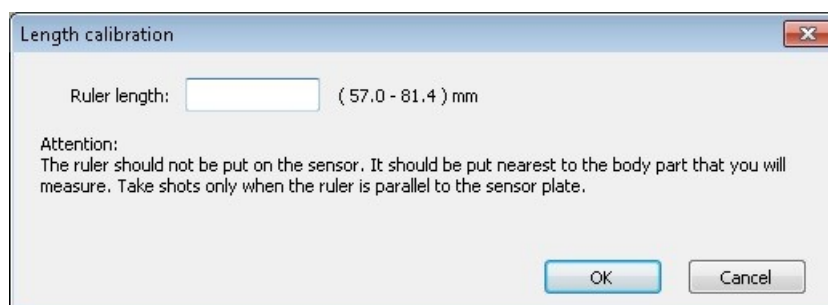
- 1 Click .
- 2 Click two specified points along the upper end vertebra.
- 3 Click two specified points along the lower end vertebra.
- 4 Click a specified point along the apical vertebra.
The Cobb angle object and the measured value are displayed.



Changing the X-Ray magnification

Based on the image captured with a ruler, changes the X-Ray magnification.

- 1 Click 
- 2 Click two specified points along the ruler.
A line and the [Length calibration] screen.
- 3 Enter the length of the ruler.
The value within the range on the screen can be entered.



- 4 Click [OK].
The X-Ray magnification is calculated. Hereafter, the modified X-Ray magnification is used for correction of the distance measurement.

NOTE

- The X-Ray magnification is deleted by returning to the study list screen or displaying another image.
- The X-Ray magnification can be selected from [X-Ray magnification] in the menu displayed by right-clicking.

Deleting the measured values or notes

To delete an object, select an object and press the Delete key.

NOTE

- Objects can be also deleted by selecting a menu displayed by right-clicking. Select [Delete] or [Delete all] from [Draw object].

Performing post-processing

Adjusting WW and WL

Adjusts the WW (Window Width) and WL (Window Level).

- 1** Click .
- 2** Point the cursor to an image and drag it.
To change the WW, drag the image to horizontal direction.
To change the WL, drag the image to vertical direction.



NOTE

- The WW and WL returns to the initial settings by returning to the study list screen or displaying another image.
- To return the WW and WL to the initial settings, select [Reset WW/WL] from [Reset] in the menu displayed by right-clicking.

Flipping images

Select [Left/Right] or [Up/Down] from [Image flip] in the menu displayed by right-clicking.

Rotating images

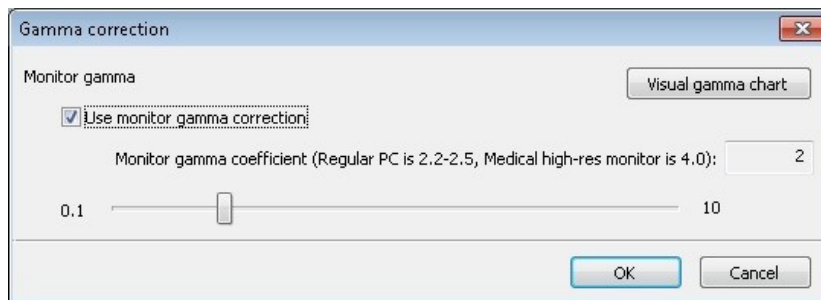
Select [Right rotation] or [Left rotation] from [Image rotation] in the menu displayed by right-clicking.

Inverting grayscale

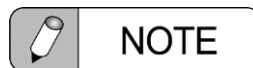
Select [Grayscale inversion] in the menu displayed by right-clicking.

Adjusting gamma

- 1 Select [Gamma correction] in the menu displayed by right-clicking.
The [Gamma correction] screen is displayed.
- 2 Move the slider.



- 3 Click [OK].



- Clear the check box of [Use monitor gamma correction] if gamma correction is not performed.
- Clicking [Visual Gamma Chart] displays the [Visual Gamma Chart] screen. Referring to the gamma chart, decide the monitor gamma coefficient.

Other operations

Printing images

- 1 Select the printing contents from [Print] in the menu displayed by right-clicking.
 - [Print at max resolution...]: Prints an image as it is displayed.
 - [Print original image...]: Prints an image in initial status. However, the result of the gamma adjustment is reflected.
- 2 Select a printer and click [Print].

Returning images to initial settings

Select [Reset all] from the [Reset] in the menu displayed by right-clicking.

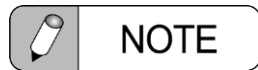
The change of image display, arrangement of objects and image processing results are deleted. However, change of X-Ray magnification and result of gamma adjustment are saved.

Playback dynamic image

Dynamic image playback tool bar is displayed for dynamic images.



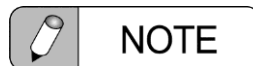
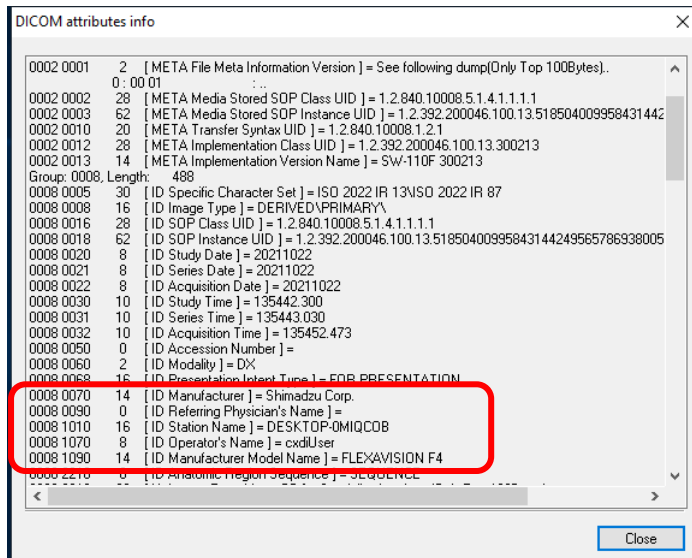
- 1 Click .
The dynamic image starts to playback.
- 2 Click .
The dynamic image pauses.



- To stop the dynamic image, click .
- To play dynamic image repeatedly, click .
- To change the frame to be displayed, move the slider in [Frame] or enter specified value and press the Enter key.
- To change the playback speed, move the slider in [Speed].

Displaying DICOM attributes information

Select [DICOM attributes info...] in the menu displayed by right-clicking.



You can confirm that the image was acquired with FLEXAVISION system from the following DICOM attribute information.

- 0008 0070 14 [ID Manufacturer] = Shimadzu Corp.
- 0008 1090 8 [ID Manufacturer Model Name] = FLEXAVISION F4